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Systems and methods for a slitter assembly

Abstract

Systems and methods for slitting a delivery device are described. A slitter assembly for slitting the delivery device in accordance with the present disclosure includes a housing, a lock mechanism coupled to the housing, and a clamshell mechanism coupled to the lock mechanism. The clamshell mechanism defines a channel having an adjustable diameter and sized to receive a device. The slitter assembly also includes a blade configured to slit a tubular shaft of the delivery device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/467,214 filed on May 17, 2023, which is incorporated by reference herein in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to a slitter assembly. In particular, the present disclosure relates to systems and methods for a slitter assembly used to slit tubular bodies.

BACKGROUND

(2) Catheter or sheath systems are widely used in the medical industry for various procedures. Such systems include tubular bodies typically made from materials such as silicone, polyurethane, or nylon. In some cases, it is necessary to split the tubular bodies lengthwise in order to expose their corresponding lumens and enable the removal of the tubular bodies from a patient. For example, catheter and sheath systems may be utilized to deliver a left ventricular (LV) lead of a cardiac resynchronization therapy (CRT) system into the coronary sinus of a patient for implantation. In order to enable the tubular body to be removed from the implanted lead and from within the patient

once implantation is complete, the tubular body may be longitudinally slit. To facilitate the slitting/splitting of a tubular body, several medical systems offer separate dedicated slitting/splitting tools. In use, these tools are aligned with and placed onto the lead after a hemostasis valve at the proximal end of the tubular body is manually split. The slitting/splitting tool is then slid over the lead until the tool's tip encounters the tubular body. The tool is then secured to the lead and the tubular body is withdrawn from the patient and over the tool, thereby slitting/splitting the tubular body and enabling the tubular body to be removed off of the lead without inadvertent lead dislodgement.

(3) Because the aforementioned slitting/splitting tools are aligned with, placed on, and secured to the lead extending through the tubular body, these tools may make it difficult to remove the tubular body. Accordingly, systems and methods for improved slitting/splitting a tubular body of a catheter or sheath are desirable.

BRIEF SUMMARY OF THE DISCLOSURE

(4) In one aspect, the present disclosure is directed to a slitter assembly including a housing, a lock mechanism coupled to the housing, and a clamshell mechanism coupled to the lock mechanism. The clamshell mechanism defines a channel having an adjustable diameter and sized to receive a device. The slitter assembly also includes a blade configured to slit a tubular shaft of a delivery device.

(5) In another aspect, the present disclosure is directed to a method for slitting a delivery device. The method includes capturing a device within a channel of a clamshell mechanism of a slitter assembly. At least a portion of the device is held in a tubular shaft of the delivery device. The method also includes pulling the delivery device through a blade of the slitter assembly while the slitter assembly and the device are stationary within the clamshell mechanism.

(6) The foregoing and other aspects, features, details, utilities, and advantages of the present disclosure will be apparent from reading the following description and claims, and from reviewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a medical device system including a slitter assembly that may be used for delivering an implantable medical device (IMD) into a patient's body.

(2) FIG. 2A is a side view of one embodiment of a slitter assembly.

(3) FIG. 2B is a sectional view of the slitter assembly of FIG. 2A.

(4) FIG. 3A is a perspective view of the slitter assembly of FIG. 2B in a first configuration.

(5) FIG. 3B is a perspective view of the slitter assembly of FIG. 2B in a second configuration.

(6) FIG. 4 is a perspective view of a clamshell mechanism of the slitter assembly of FIG. 2A.

(7) FIG. 5A is a perspective view of a lock mechanism of the slitter assembly of FIG. 2A.

(8) FIG. 5B is another perspective view of a lock mechanism of the slitter assembly of FIG. 2A.

(9) FIG. 6 is a sectional view of a slitter of the slitter assembly of FIG. 2A.

(10) FIGS. 7A-7C are schematic diagrams of one embodiment of the slitter assembly of FIG. 2A interacting with a delivery device.

(11) FIG. 8 is flow diagram of a method of slitting a delivery device.

DETAILED DESCRIPTION OF THE DISCLOSURE

(12) The present disclosure provides systems, devices, and methods for a slitter assembly used to slit tubular bodies. The slitter assembly in accordance with the present disclosure includes a blade and a clamshell mechanism coupled to the slitter assembly. The slitter assembly also includes a ramp and a lock mechanism coupled to the clamshell mechanism and the ramp. The lock mechanism includes a lever, at least one tab, and a lock feature, and is configured to expand and

collapse the clamshell mechanism.

(13) It is generally known that various medical procedures, using catheter and sheath systems, require that the tubular bodies of these systems be split (e.g., to enable removal of the tubular bodies from a patient's body). To split the tubular bodies, slitting/splitting tools may be used. However, at least some known splitting/splitting tools must be aligned with, placed on, and secured to a lead extending through a lumen of the tubular body, which may make the tubular body removal process difficult.

(14) As used herein, the terms “proximal” and “distal,” refer to a direction relative to a user of a delivery device. “Proximal” refers to a direction toward the end of the delivery device near to the user, and “distal” refers to a direction away from the user and (generally) inside the body of a patient. As used herein, the terms “substantially,” “generally,” and “about” are intended to mean that slight deviations from absolute are included within the scope of the term so modified.

(15) FIG. 1 is a schematic diagram of a medical device system **100** used for delivering an implantable medical device (IMD) into a patient's body. In general, system **100** includes a slitter assembly **102** and a delivery device **104** (e.g., a catheter or sheath) configured to deliver an implantable device **106** into a patient's body. Device **106** may include an implantable medical device such as an implantable medical lead, an inner catheter or outer sheath, an introducer, a stylet, a guidewire, a sensor, or other accessories or devices typically delivered via delivery device **104**. Delivery device **104** may have a tubular shaft **108** and a hub **110** on a proximal end **112** of delivery device **104**. Tubular shaft **108** may include a lumen **114** extending the length of tubular shaft **108**. Lumen **114** provides a passageway for device **106** to enter the body, for example, the patient's heart **116** during implantation of device **106** (e.g., a lead). Upon placement of device **106**, delivery device **104** may be removed from device **106** by slitting delivery device **104**, and in particular tubular shaft **108**, along its length.

(16) In one embodiment, tubular shaft **108** is integrated into hub **110** such that an inner wall of tubular shaft **108** forms a longitudinally extending strip of a wall **118** of hub **110**. The longitudinal length of wall **118** extends from a proximal end **120** of hub **110** to a distal end **122** of hub **110**. Accordingly, the material forming tubular shaft **108** may extend the length of delivery device **104** uninterrupted and fully accessible to a slitting tool, such as slitter assembly **102**, through the length of hub **110**. When using slitter assembly **102** to slit delivery device **104** along its length, including the lengths of tubular shaft **108** and hub **110**, slitter assembly **102** may encounter material of tubular shaft **108** without encountering material of hub **110**, thereby enabling delivery device **104** to be slit and removed from the device **106** without disrupting device **106**. For example, delivery device may be slit and removed from device **106** while device **106** remains stationary. Hub **110** includes a hemostasis valve **124** and a slitting path **126** (e.g., an integrated longitudinally extending shaft strip) that reduces a hub-to-shaft transitional jerk that may occur, thereby reducing complications associated with dislodging a placed lead or other device **106**, such as increased procedure time or damage to cardiac tissue. A cap **128** and hemostasis valve **124** may be coupled to hub **110**, and cap **128** may retain hemostasis valve **124** within hub **110**.

(17) During implantation of device **106**, tubular shaft **108** may be inserted into the patient's heart **116** via the subclavian vein **130**, as shown. Alternatively, and in other embodiments, the tubular shaft **108** may be inserted into the patient's heart **116** via a femoral vein (not shown) or other suitable entry point to the patient's body. Once tubular shaft **108** is in position, device **106** may be inserted therethrough. For example, once tubular shaft **108** is in position, device **106**, such as a lead, may be inserted through hemostasis valve **124** in hub **110** and through lumen **114** of tubular shaft **108** so that a distal tip **132** (e.g., a lead tip) at the distal end of the device **106** may be guided into position in the patient's heart **116**.

(18) Device **106** includes a proximal end **134** and a distal end **136**. In one embodiment, proximal end **134** includes an electrical connector **138** for mechanically and electrically coupling proximal end **134** to an IMD, such as a pulse generator (e.g., pacemakers, defibrillators, or implantable

cardioverter defibrillators (ICD)). Electrical connector **138** is sized to prevent delivery device **104** from being proximally withdrawn from device **106**. Once device **106** is implanted or placed into position, as appropriate, delivery device **104** may be slit to enable delivery device **104** to clear electrical connector **138** or proximal end **134** as delivery device **104** is removed from device **106**. (19) As mentioned above, the material forming tubular shaft **108** extends into hub **110** to form at least a longitudinal strip of wall **118** of hub **110**. Therefore, slitting path **126** for slitting delivery device **104**, including tubular shaft **108** and hub **110**, extends along shaft material and does not encounter hub material, or at least any significant amount of hub material. With respect to slitting path **126**, delivery device **104** has no hub-to-shaft transition. In other words, because tubular shaft **108** is integrated into hub **110**, delivery device **104** does not require removal of hub **110** in order to slit tubular shaft **108**.

(20) Once device **106** is implanted or placed at the desired location, delivery device **104** may be slit with slitter assembly **102**. As discussed above, cap **128** and hemostasis valve **124** may be coupled to hub **110**. In some embodiments, an arcuate opening **140** in cap **128** is configured to enable passage of a slitter blade of slitter assembly **102** through cap **128** and hemostasis valve **124** such that removal of the cap **128** prior to slitting is not required. Also, hemostasis valve **124** may be slit while in place within hub **110**. Slitter assembly **102** is configured to (a) facilitate capturing of device **106** using a clamshell mechanism of slitter assembly **102**, (b) cover and protect device **106** during the slitting process while delivery device **104** is separated from device **106**, and (c) easily slit delivery device **104** by providing a slitter blade including an angle ranging between 10 and 50 degrees that may reduce the force used to slit delivery device **104**. As a result, slitter assembly **102** enables slitting with low and consistent slit forces along the length of delivery device **104**, substantially reducing, if not completely eliminating, the transition jerk. The possibility of dislodging or disrupting the position of device **106** is thereby reduced or eliminated.

(21) FIG. 2A is a side view of one embodiment of slitter assembly **102**. Slitter assembly **102** includes a housing **200**, a distal section **202** and a proximal section **204**. Distal section **202** includes a blade **206** and a clamshell mechanism **208** coupled to blade **206**. Clamshell mechanism **208** may be made of plastic material, such as Acrylonitrile Butadiene Styrene (ABS) plastic, to enable secure capturing of device **106**, as described herein. Clamshell mechanism **208** defines a channel **210** covering a portion of blade **206** and a portion of device **106** (shown in FIG. 1). When using slitter assembly **102** to slit delivery device **104** (shown in FIG. 1) along its length, device **106** is introduced into channel **210**, such that blade **206** may slit the lengths of tubular shaft **108** and hub **110** (both shown in FIG. 1) and delivery device **104** may be removed from the device **106** without disrupting device **106**.

(22) Distal section **202** also includes a lock mechanism **212** coupled to the clamshell mechanism. Lock mechanism **212** includes a lever **214**, and is configured to selectively expand and collapse clamshell mechanism **208**, in particular channel **210**. By collapsing and expanding clamshell mechanism **208**, different diameter sizes of device **106** may be accommodated within channel **210**. In other words, the diameter of channel **210** is adjustable. The outer diameter size of channel **210** may be range between about 0.120 inches (0.30 cm) in a closed position and about 0.200 inches (0.51 cm) in an open position. The inner diameter of channel **210** may be in an oval shape that may clamp or receive up to a 6 French (about 2 mm) lead. The inner diameter size of channel **210** may range between about 0.060 inches (0.15 cm) in a closed position and 0.138 inches (0.35 cm) in an open position. In other words, when clamshell mechanism **208** is closed, the inner diameter of channel **210** may be about 60 thousandth of an inch (about 0.15 cm), thereby enabling channel **210** to clamp a 4 French (about 0.13 cm) lead. As a result, slitter assembly **102** provides an active capture feature to securely hold and encapsulate different diameters of device **106** during the slitting process, thereby reducing, if not completely eliminating, the transition jerk, damage to proximal and distal ends of delivery device **104**, and the possibility of dislodging or disrupting the position of device **106**.

(23) Distal section **202** of housing **200** further includes a thumb location **216** that guides a clinician to hold slitter assembly **102** with either hand of the clinician, while placing the clinician's thumb on thumb location **216**. This hand position encourages axial alignment with delivery device **104** during the slitting process.

(24) FIG. 2B is a sectional view of slitter assembly **102** shown in FIG. 2A. In addition to the components of slitter assembly **102** described in FIG. 2A, slitter assembly **102** includes a ramp **218** coupled to lock mechanism **212**. Ramp **218** activates (e.g., moves or translates) in response to lever **214** changing its position. In other words, ramp **218** is driven by the position (e.g., up, down) of lever **214**. By changing the position of lever **214**, a portion of lock mechanism **212** pivots within the interior of housing **200**, causing a far end of lock mechanism **212** to pivot about a housing pin **220**. The end of the lock mechanism **212** is shaped so as to engage an end of ramp mechanism **218**, causing ramp mechanism **218** to translate linearly within housing **200** to open/close clamshell mechanism **208**.

(25) When ramp **218** activates, clamshell mechanism **208** moves from a first configuration to a second configuration. For example, when lever **214** is in a down position as further shown in FIG. 3A, clamshell mechanism **208** is open. In another example, when lever **214** is in an up position as shown further in FIG. 3B, clamshell mechanism **208** is closed. In general, when lever **214** is down and clamshell mechanism **208** is open, device **106** (shown in FIG. 1) may be introduced into channel **210**. Once device **106** is placed in channel **210**, lever **214** may be moved up so that clamshell mechanism **208** starts closing until clamshell mechanism **208** stops closing, thereby capturing device **106** (e.g., capturing device **106** at about the diameter of device **106**). At this point with device **106** contained within closed channel **210**, a clinician may start slitting delivery device **104** to remove delivery device **104** from device **106**. Since device **106** is held firm within channel **210** of clamshell mechanism **208**, the possibility of dislodging or disrupting the position of device **106** is reduced or eliminated during the slitting process. Upon removal of delivery device **104**, lever **214** may be moved to the down position to open clamshell mechanism **208** and remove device **106** from slitter assembly **102**.

(26) FIG. 3A is a sectional view of slitter assembly **102** shown in FIG. 2B in a first configuration. In the first configuration, lever **214** is down and clamshell mechanism **208**, and in particular, channel **210**, is open. Clamshell mechanism **208** includes a top side **302** and a bottom side **304** forming channel **210**. Top side **302** is positioned opposite to bottom side **304**. When channel **210** moves from an open position to a closed position, or vice versa, only one of top side **302** or bottom side **304** moves to open or close channel **210**. For example, if top side **302** is movable, then bottom side **304** is stationary, and vice versa. As mentioned above, lock mechanism **212** is coupled to a ramp **218**, such that ramp **218** activates in response to lever **214** changing its position. When ramp **218** activates, clamshell mechanism **208** moves from the first configuration (e.g., open) to a second configuration (e.g., closed). In particular, ramp **218** moves the movable side of channel **210**. In other words, ramp **218** moves one of top side **302** or bottom side **304**, whichever is coupled to ramp **218**, to close channel **210**. In general, channel **210** moves from the open configuration to the closed configuration once device **106** (shown in FIG. 1) is placed into channel **210**.

(27) FIG. 3B is a sectional view of slitter assembly **102** shown in FIG. 2B in a second configuration. In the second configuration, lever **214** is up and clamshell mechanism **208** is closed, in particular channel **210**. As described above, only one of top side **302** or bottom side **304** is movable. When ramp **218** activates, clamshell mechanism **208** moves from the second configuration (e.g., closed) to the first configuration (e.g., open). In particular, ramp **218** moves the movable side of channel **210**. In other words, ramp **218** moves one of top side **302** or bottom side **304**, whichever is coupled to ramp **218**, to open channel **210**. In general, channel **210** moves from the open configuration to the closed configuration once device **106** (shown in FIG. 1) is placed into channel **210**.

(28) FIG. 4 is a perspective view of clamshell mechanism **208** of slitter assembly **102**. As described

above, when ramp **218** activates, clamshell mechanism **208** moves from one configuration to another (e.g., from an open configuration to a closed configuration, or vice versa). In one embodiment, bottom side **304** is movable and top side **302** is stationary, where ramp **218** is coupled to bottom side **304** and is located under bottom side **304**. Bottom side **304** includes a tab **402** coupled to ramp **218** and top side **302**. In this embodiment, when ramp **218** translates downwardly within housing **200**, tab **402** pivots about a pivot pin (not shown) connected to housing **200**, pushing tab **402** towards the stationary portion (e.g., bottom side **304**) of clamshell mechanism **208** and causing clamshell mechanism **208** to actuate.

(29) In one example, when ramp **218** activates, ramp **218** moves away (e.g., retracts) from bottom side **304**. By moving away from bottom side **304**, ramp **218** releases a portion of bottom side **304**, including tab **402**, thereby causing channel **210** to open. In another example, when ramp **218** activates, ramp **218** moves toward bottom side **304**, such that ramp **218** supports a portion of bottom side **304**, including tab **402**, thereby causing channel **210** to close. In some embodiments, a spring element may be coupled to ramp **218** and bottom side **304** of clamshell mechanism **208**. The spring element may bias ramp **218** and may assist in pushing ramp **218** back when lever **214** is moved to a down position. In other embodiments, a release mechanism may be coupled to ramp **218** and bottom side **304**, and perform similar functions to those of tab **402**, as described herein. In yet other embodiments, other mechanisms may be used in combination with ramp **218** to enable channel **210** to open and close, as described herein.

(30) FIGS. 5A and 5B are perspective views of lock mechanism **212** of slit assembly **102**. In addition to lever **214**, lock mechanism **212** includes a lock tab **502** and a lock feature **504**. In one embodiment, lock feature **504** may include a V lock feature, which may be a mechanism that uses low force to flex up a plastic feature (e.g., lock tab **502** and/or lock feature **504**) during movement, and flex back down the plastic feature into grooves (e.g., lock grooves **506** described in detail below) that restrict movement. In particular, lock feature **504** may include a bendable tab (e.g., the V lock feature) that pivots at its base to permit a detent feature that engages with a corresponding lock groove **506** when lock mechanism **212** is in either a fully open (e.g., lever **214** is in the down direction) or fully closed position (e.g., lever **214** is in the up direction). A force in either the up or down direction (depending on whether lever **214** is in the open or closed position) causes the bendable tab to deflect slightly and enables a detent of lock feature **504** to disengage from the corresponding lock groove **506**, thereby allowing lever **214** and lock mechanism **212** to move to another position. Lock feature **504** may be made of flexible material that enables engaging and disengaging from lock grooves **506**. Lock feature **504** is configured to bend at its base and displace to permit the detent to disengage from lock groove **506** by applying low force to flex up the plastic feature (e.g., tab **502** and/or lock feature **504**) during movement. In other embodiments, lock feature **504** may include other suitable lock features that enable lock mechanism **212** to function, as described herein.

(31) FIG. 5A shows a top view of lock mechanism **212**, and FIG. 5B shows a bottom view of lock mechanism **212**. In one embodiment, lock feature **504** locks lever **214** at a desired position by engaging with one of a plurality of lock grooves **506** of slit assembly **102**. Lock grooves **506** are positioned under lock tab **502** and lock feature **504**. For example, when lever **214** is changed from an open configuration (shown in FIG. 4) to a closed configuration, lock feature **504** may disengage from a lock groove **506** for opening channel **210**, move to another lock groove **506**, and engage in the other lock groove **506**, causing lock mechanism **212** to close channel **210**. In one embodiment, the plurality of lock grooves **506** includes two lock grooves **506** (e.g., one lock groove **506** for opening channel **210** and another lock groove **506** for closing channel **210**). In other embodiments, the plurality of lock grooves **506** includes more than two lock grooves **506**, where each lock groove **506** may specify a particular diameter for channel **210**.

(32) FIG. 6 is a sectional view of an example blade **206** of slit assembly **102** (shown in FIG. 2A). As described above, blade **206** is coupled to clamshell mechanism **208** (shown in FIG. 2A),

where a portion of clamshell mechanism **208** includes channel **210** (shown in FIG. 2A) covering a portion of blade **206** and a portion of device **106** (shown in FIG. 1). In one embodiment, the cutting edge of blade **206** includes an angle α that may range between 10 and 50 degrees (e.g., angle α may be 30 degrees) with respect to a horizontal base of blade **206**. Angle α enables a shallower entry of delivery device **104**, in particular tubular shaft **108** (both shown in FIG. 1), in comparison to conventional blades, which include an angle of about 60 degrees. A shallow entry of delivery device **104** permits easily slitting delivery device **104** by reducing the force required to slit delivery device **104**.

(33) FIGS. 7A-7C are schematic diagrams of one embodiment of slitter assembly **102** interacting with delivery device **104**. As shown in FIG. 7A, the slitting process may be initiated by introducing device **106** into clamshell mechanism **208**. Clamshell mechanism **208** may partially cover device **106** and blade **206**. In particular, device **106** is introduced into channel **210** of clamshell mechanism **208**. Device **106** is also introduced into arcuate opening **140** (shown in FIG. 1) in distal end **122** of hub **110** of delivery device **104**, such that device **106** may be introduced into tubular shaft **108** of delivery device **104**. Device **106** may be in contact with blade **206**. When using slitter assembly **102** to slit delivery device **104** along its length, device **106** is introduced into channel **210** and the opening in distal end **122** of hub **110**, such that blade **206** may slit the lengths of tubular shaft **108** and hub **110** and delivery device **104** may be removed from the device **106** without disrupting device **106**.

(34) FIG. 7B shows slitter assembly **102** in a subsequent step to the initiation of the slitting process shown in FIG. 7A. In FIG. 7B, clamshell mechanism **208** and channel **210** are introduced into delivery device **104**. Specifically, device **106** and blade **206** (both shown in FIG. 7A) are introduced into arcuate opening **140** (shown in FIG. 1) in hub **110** and tubular shaft **108**. During this subsequent step, lock mechanism **212** and lever **214** are in a lock position to securely hold and encapsulate device **106** within channel **210** during the slitting process.

(35) FIG. 7C shows slitter assembly **102** in a step subsequent to the step shown in FIG. 7B. In FIG. 7C, delivery device **104** is pulled through blade **206** while slitter assembly **102** and device **106** (shown in FIG. 7A) remain stationary during the slitting process. Similar to FIG. 7B, lock mechanism **212** and lever **214** are in a lock position to securely hold and encapsulate device **106** within channel **210**. As described above, blade **206**, positioned in clamshell mechanism **208**, slits tubular shaft **108** along the length of tubular shaft **108** to remove tubular shaft **108** and delivery device **104** from device **106** (shown in FIG. 7A). By doing so, the slitting process substantially reduces, if not completely eliminates, the transition jerk. The possibility of dislodging or disrupting the position of device **106** is thereby reduced or eliminated.

(36) FIG. 8 is flow diagram of a method **800** of slitting a delivery device **104** (shown in FIG. 1). Method **800** may be implemented, for example, by slitter assembly **102** (shown in FIG. 2A). Method **800** includes capturing **802** device **106** within channel **210** of clamshell mechanism **208** (both shown in FIG. 7A) of slitter assembly **102**. At least a portion of device **106** is held in tubular shaft **108** (shown in FIG. 7A) of delivery device **104**. As described herein, the diameter of channel **210** is adjustable, enabling different sizes (e.g., diameter sizes) of device **106** to be captured. The sizes of device **106** may vary as device **106** may include an implantable medical lead, an inner catheter or outer sheath, an introducer, a stylet, a guidewire, a sensor, or other accessories or devices typically delivered via delivery device **104**, as described herein.

(37) Method **800** also includes pulling **804** delivery device **104** through blade **206** (shown in FIG. 7A) of slitter assembly **102** while slitter assembly **102** and device **106** are stationary. Pulling **804** may include, for example, slitting tubular shaft **108** along a length of tubular shaft **108** to remove device **106** from delivery device **104**.

(38) It should be understood that method **800** may include additional and/or alternative steps to those set forth above. For example, in some embodiments, method **800** may also include activating ramp **218** when lever **214** (both shown in FIG. 3A) transitions between different position, and in

response to activating ramp 218, method 800 further includes opening or closing channel 210 of clamshell mechanism 208, or causing, by lock tab 502 (shown in FIG. 502), one side of channel 210 to open or close channel 210. In addition, in other embodiments, method 800 may include receiving, a lock groove 506 (shown in FIG. 7A), lock feature 504 (shown in FIG. 7B), and locking channel 210 in an open configuration or a closed configuration. In yet other embodiments, method 800 may include moving, using lever 214, lock feature 504 from one lock groove 506 to another lock groove 506.

(39) Although certain embodiments of this disclosure have been described above with a certain degree of particularity, those skilled in the art could make numerous alterations to the disclosed embodiments without departing from the spirit or scope of this disclosure. All directional references (e.g., upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of the disclosure. Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements and relative movement between elements. As such, joinder references do not necessarily infer that two elements are directly connected and in fixed relation to each other. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the spirit of the disclosure as defined in the appended claims.

(40) When introducing elements of the present disclosure or the preferred embodiment(s) thereof, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(41) As various changes could be made in the above constructions without departing from the scope of the disclosure, it is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

Claims

1. A slitter assembly comprising: a housing; a lock mechanism coupled to the housing, wherein the housing comprises a plurality of lock grooves, each lock groove configured to receive a lock feature of the lock mechanism; a clamshell mechanism coupled to the lock mechanism, wherein the clamshell mechanism defines a channel having an adjustable diameter and sized to receive a device, and wherein the lock mechanism is configured to lock the channel in an open configuration or a closed configuration depending upon which one of the lock grooves engages with the lock feature; and a blade configured to slit a tubular shaft of a delivery device.
2. The slitter assembly of claim 1, wherein the lock mechanism comprises a lever and a ramp, and wherein the ramp is configured to be activated when the lever transitions between different positions.
3. The slitter assembly of claim 2, wherein the ramp is coupled to the clamshell mechanism, and wherein when the ramp is activated, the clamshell mechanism opens or closes the channel.
4. The slitter assembly of claim 2, wherein the clamshell mechanism comprises a lock tab coupled to the ramp and one of two sides of the channel, and wherein when the ramp is activated, the lock tab causes the one side of the channel to open or close the channel.
5. The slitter assembly of claim 1, wherein an outer diameter of the channel ranges between 0.51 cm in the open configuration and 0.30 cm in the closed configuration.
6. The slitter assembly of claim 1, wherein an inner diameter of the channel ranges between 0.35 cm in the open configuration and 0.15 cm in the closed configuration.

7. The slitter assembly of claim 1, wherein a cutting edge of the blade forms an angle greater than or equal to 10 degrees and equal to or less than 50 degrees with respect to a horizontal base of the blade.

8. The slitter assembly of claim 1, wherein the lock mechanism comprises a lever coupled to the lock feature, and wherein the lever is operable to move the lock feature from one lock groove of the plurality of lock grooves to another lock groove of the plurality of lock grooves.

9. A method for slitting a delivery device, the method comprising providing a slitter assembly including a housing, and a lock mechanism coupled to the housing, wherein the housing includes a plurality of lock grooves, each lock groove configured to receive a lock feature of the lock mechanism; capturing a device within a channel of a clamshell mechanism of the slitter assembly, wherein at least a portion of the device is held in a tubular shaft of the delivery device, and wherein the channel has an adjustable diameter and sized to receive the device, and wherein the capturing comprises (a) receiving, at one of the plurality of lock grooves, the lock feature and (b) locking the channel in an open configuration or a closed configuration using the lock feature; and pulling the delivery device through a blade of the slitter assembly while the slitter assembly and the device are stationary, wherein the blade is configured to slit the tubular shaft of the delivery device.

10. The method of claim 9, wherein pulling the delivery device through the blade further comprises slitting the tubular shaft along a length of the tubular shaft to remove the device from the delivery device.

11. The method of claim 9, wherein the a lock mechanism includes a lever and a ramp, and wherein the ramp is coupled to the clamshell mechanism, and wherein the method further comprises: activating the ramp when the lever transitions between different positions; and in response to activating the ramp, opening or closing the channel of the clamshell mechanism.

12. The method of claim 11, wherein the clamshell mechanism includes a lock tab coupled to the ramp and one of two sides of the channel, and wherein the method further comprises, in response to activating the ramp, causing, by the lock tab, the one side of the channel to open or close the channel.

13. The method of claim 9, further comprising providing the channel to have an outer diameter range between 0.51 cm in an open configuration and 0.30 cm in a closed configuration, and providing the channel to have an inner diameter range between 0.35 cm in the open configuration and 0.15 cm in the closed configuration.

14. The method of claim 9, further comprising providing the blade with a cutting edge having an angle greater than or equal to 10 degrees and equal to or less than 50 degrees with respect to a horizontal base of the blade.

15. The method of claim 9, further comprising providing the delivery device to include a hub having an arcuate opening configured to receive the clamshell mechanism.

16. The method of claim 9, wherein the lock mechanism includes a lever coupled to the lock feature, and wherein the method further comprises moving, using the lever, the lock feature from one lock groove of the plurality of lock grooves to another lock groove of the plurality of lock grooves.
